

SharedFileOut.h

Change log

- [2026-03-31] inner structures have fixed size (see the section structure name [size])
- [2026-03-31] **!!!** display_current_page_index is now an array of 16 items (it was 9 before)
- [2026-03-31] the example solution has been updated

Enumerations

ACEVO_STATUS

Current operational state of the simulator.

Value	Meaning
AC_OFF (0)	Simulator is not running / no session active
AC_REPLAY (1)	A replay is currently being played back
AC_LIVE (2)	Live driving session is active
AC_PAUSE (3)	Session is paused

ACEVO_SESSION_TYPE

Type of racing session currently loaded.

Value	Meaning
AC_UNKNOWN (-1)	Session type not yet determined
AC_TIME_ATTACK (0)	Time attack / qualifying session
AC_RACE (1)	Race session
AC_HOT_STINT (2)	Hot-stint practice
AC_CRUISE (3)	Untimed cruise

ACEVO_FLAG_TYPE

Race flag currently shown to the driver.

Value	Meaning
AC_NO_FLAG (0)	No flag displayed
AC_WHITE_FLAG (1)	Slow vehicle ahead on track
AC_GREEN_FLAG (2)	Track clear — racing resumed
AC_RED_FLAG (3)	Session stopped due to incident or hazard
AC_BLUE_FLAG (4)	Lapped car must yield to the race leader
AC_YELLOW_FLAG (5)	Hazard present — no overtaking
AC_BLACK_FLAG (6)	Driver disqualified / must pit immediately
AC_BLACK_WHITE_FLAG (7)	Warning for unsportsmanlike behaviour
AC_CHECKERED_FLAG (8)	Session or race has ended
AC_ORANGE_CIRCLE_FLAG (9)	Mechanical problem — car must pit
AC_RED_YELLOW_STRIPES_FLAG (10)	Slippery surface ahead on track

ACEVO_CAR_LOCATION

Where on the circuit the car is currently positioned.

Value	Meaning
ACEVO_UNASSIGNED (0)	Position not yet determined
ACEVO_PITLANE (1)	Car is inside the pit lane
ACEVO_PITENTRY (2)	Car is at the pit-lane entry
ACEVO_PITEXIT (3)	Car is at the pit-lane exit
ACEVO_TRACK (4)	Car is on the racing circuit

ACEVO_ENGINE_TYPE

Powertrain type of the player car.

Value	Meaning
ACEVO_INTERNAL_COMBUSTION (0)	Traditional petrol/diesel internal combustion engine
ACEVO_ELECTRIC_MOTOR (1)	Fully electric powertrain

ACEVO_STARTING_GRIP

Initial grip conditions at session start.

Value	Meaning
ACEVO_GREEN (0)	Track grip at minimum
ACEVO_FAST (1)	Track grip in advanced (fast) stage
ACEVO_OPTIMUM (2)	Track conditions starting at optimum grip

Structs

SPageFilePhysics

Raw physics telemetry updated every simulation step. Contains all low-level vehicle dynamics data.

Type	Field	Meaning
int	packetId	Incrementing counter — detect new data packets by comparing to previous value
float	gas	Throttle pedal position (0.0 = released, 1.0 = full throttle)
float	brake	Brake pedal position (0.0 = released, 1.0 = full brake)
float	fuel	Remaining fuel in litres
int	gear	Engaged gear: 0 = reverse, 1 = neutral, 2+ = forward gears
int	rpms	Engine speed in revolutions per minute
float	steerAngle	Normalised steering angle (−1.0 = full left, +1.0 = full right)
float	speedKmh	Vehicle speed in km/h
float[3]	velocity	World-space velocity vector [X, Y, Z] in m/s
float[3]	accG	Acceleration in G [lateral X, longitudinal Y, vertical Z]
float[4]	wheelSlip	Tyre slip value per wheel [FL, FR, RL, RR]
float[4]	wheelLoad	Vertical tyre load in Newtons [FL, FR, RL, RR]
float[4]	wheelsPressure	Tyre inflation pressure in PSI [FL, FR, RL, RR]
float[4]	wheelAngularSpeed	Wheel rotational speed in rad/s [FL, FR, RL, RR]
float[4]	tyreWear	Tyre wear level (0.0 = new, 1.0 = fully worn) [FL, FR, RL, RR]
float[4]	tyreDirtyLevel	Amount of dirt / debris on each tyre surface [FL, FR, RL, RR]
float[4]	tyreCoreTemperature	Core temperature of each tyre in °C [FL, FR, RL, RR]
float[4]	camberRAD	Wheel camber angle in radians per corner [FL, FR, RL, RR]
float[4]	suspensionTravel	Suspension compression travel in metres [FL, FR, RL, RR]
float	drs	DRS flap state (0.0 = closed, 1.0 = fully open)
float	tc	Traction control cut intensity (0.0 = inactive, 1.0 = maximum)
float	heading	Vehicle heading relative to world north in radians
float	pitch	Chassis pitch angle in radians (positive = nose up)
float	roll	Chassis roll angle in radians (positive = right side down)
float	cgHeight	Height of the centre of gravity above the ground in metres
float[5]	carDamage	Damage level per body zone [front, rear, left, right, centre] (0.0–1.0)
int	numberOfTyresOut	Number of tyres currently outside track limits
int	pitLimiterOn	Pit-speed limiter active (0 = off, 1 = on)
float	abs	ABS intervention intensity (0.0 = inactive, 1.0 = fully active)
float	kersCharge	KERS/ERS battery state of charge (0.0–1.0)
float	kersInput	KERS/ERS power delivery level currently being deployed (0.0–1.0)
int	autoShifterOn	Automatic gearshift aid active (0 = manual, 1 = auto)
float[2]	rideHeight	Ride height at front and rear axle in metres [front, rear]
float	turboBoost	Current turbo boost pressure in bar
float	ballast	Additional ballast added to the car in kg
float	airDensity	Ambient air density in kg/m ³
float	airTemp	Ambient air temperature in °C
float	roadTemp	Road surface temperature in °C

Type	Field	Meaning
float[3]	localAngularVel	Angular velocity in the car's local frame [pitch, yaw, roll] in rad/s
float	finalFF	Final force-feedback torque value sent to the wheel (Nm)
float	performanceMeter	Real-time delta vs. best lap (positive = ahead of reference)
int	engineBrake	Engine-braking setting level (higher = more engine braking)
int	ersRecoveryLevel	ERS energy-recovery intensity level
int	ersPowerLevel	ERS power-deployment level
int	ersHeatCharging	ERS heat-charging mode active (0 = off, 1 = on)
int	ersIsCharging	ERS currently recovering energy (0 = deploying, 1 = charging)
float	kersCurrentKJ	Energy stored in the KERS/ERS battery in kilojoules
int	drsAvailable	DRS can be activated (0 = no, 1 = yes)
int	drsEnabled	DRS is open and active (0 = closed, 1 = open)
float[4]	brakeTemp	Brake disc temperature per corner in °C [FL, FR, RL, RR]
float	clutch	Clutch pedal position (0.0 = engaged, 1.0 = fully disengaged)
float[4]	tyreTempI	Tyre inner-edge temperature per wheel in °C [FL, FR, RL, RR]
float[4]	tyreTempM	Tyre mid-tread temperature per wheel in °C [FL, FR, RL, RR]
float[4]	tyreTempO	Tyre outer-edge temperature per wheel in °C [FL, FR, RL, RR]
int	isAIControlled	Car is driven by AI (0 = player, 1 = AI)
float[4][3]	tyreContactPoint	3-D world-space contact point of each tyre with the road [FL,FR,RL,RR][X,Y,Z]
float[4][3]	tyreContactNormal	Road-surface normal vector at each tyre contact point [FL,FR,RL,RR][X,Y,Z]
float[4][3]	tyreContactHeading	Heading vector at each tyre contact point [FL,FR,RL,RR][X,Y,Z]
float	brakeBias	Front brake-bias ratio (e.g. 0.56 = 56 % front)
float[3]	localVelocity	Velocity in the car's local reference frame [X, Y, Z] in m/s
int	P2PActivations	Remaining Push-to-Pass activations
int	P2PStatus	Push-to-Pass status (0 = inactive, 1 = active)
int	currentMaxRpm	Current rev-limiter ceiling in RPM
float[4]	mz	Self-aligning tyre torque (Mz) per wheel [FL, FR, RL, RR] in Nm
float[4]	fx	Longitudinal tyre force (Fx) per wheel [FL, FR, RL, RR] in N
float[4]	fy	Lateral tyre force (Fy) per wheel [FL, FR, RL, RR] in N
float[4]	slipRatio	Longitudinal slip ratio per tyre [FL, FR, RL, RR]
float[4]	slipAngle	Lateral slip angle per tyre in radians [FL, FR, RL, RR]
int	tcinAction	Traction control currently cutting power (0 = no, 1 = yes)
int	absInAction	ABS currently modulating brakes (0 = no, 1 = yes)
float[4]	suspensionDamage	Suspension structural damage per corner (0.0–1.0) [FL, FR, RL, RR]
float[4]	tyreTemp	Representative tyre surface temperature per wheel in °C [FL, FR, RL, RR]
float	waterTemp	Engine coolant temperature in °C
float[4]	brakeTorque	Braking torque at each wheel in Nm [FL, FR, RL, RR]
int	frontBrakeCompound	Front brake-pad compound identifier
int	rearBrakeCompound	Rear brake-pad compound identifier
float[4]	padLife	Brake-pad remaining life per corner (0.0–1.0) [FL, FR, RL, RR]
float[4]	discLife	Brake-disc remaining life per corner (0.0–1.0) [FL, FR, RL, RR]
int	ignitionOn	Ignition switch state (0 = off, 1 = on)
int	starterEngineOn	Starter motor currently cranking (0 = no, 1 = yes)

Type	Field	Meaning
int	isEngineRunning	Engine is running (0 = stopped, 1 = running)
float	kerbVibration	Vibration intensity transmitted from kerb strikes
float	slipVibrations	Vibration intensity caused by tyre slip
float	roadVibrations	Vibration intensity from road surface texture
float	absVibrations	Vibration intensity generated by ABS pulsing

SMEvoTyreState [256 bytes]

Complete state of a single tyre corner. Embedded four times in SPageFileGraphicEvo (lf, rf, lr, rr).

Type	Field	Meaning
float	slip	Combined tyre slip magnitude
bool	lock	Tyre is locked under braking (true = locking)
float	tyre_pression	Tyre inflation pressure (PSI)
float	tyre_temperature_c	Average tyre carcass temperature in °C
float	brake_temperature_c	Brake disc temperature in °C
float	brake_pressure	Hydraulic brake pressure applied at this corner
float	tyre_temperature_left	Inner-edge tyre temperature in °C
float	tyre_temperature_center	Centre-tread tyre temperature in °C
float	tyre_temperature_right	Outer-edge tyre temperature in °C
char[33]	tyre_compound_front	Name of the compound fitted on the front axle
char[33]	tyre_compound_rear	Name of the compound fitted on the rear axle
float	tyre_normalized_pressure	Pressure as a 0–1 fraction of the target range
float	tyre_normalized_temperature_left	Inner-edge temperature as a 0–1 fraction of optimal range
float	tyre_normalized_temperature_center	Centre temperature as a 0–1 fraction of optimal range
float	tyre_normalized_temperature_right	Outer-edge temperature as a 0–1 fraction of optimal range
float	brake_normalized_temperature	Brake temperature as a 0–1 fraction of optimal operating range
float	tyre_normalized_temperature_core	Core tyre temperature as a 0–1 fraction of optimal range

SMEvoDamageState [128 bytes]

Structural damage level for each body zone of the car (0.0 = undamaged, 1.0 = destroyed).

Type	Field	Meaning
float	damage_front	Damage on the front body / nose
float	damage_rear	Damage on the rear body / diffuser
float	damage_left	Damage on the left side of the body
float	damage_right	Damage on the right side of the body
float	damage_center	Damage on the central / underfloor area
float	damage_suspension_lf	Damage on the front-left suspension
float	damage_suspension_rf	Damage on the front-right suspension
float	damage_suspension_lr	Damage on the rear-left suspension
float	damage_suspension_rr	Damage on the rear-right suspension

SMEvoPitInfo [64 bytes]

Status of each pit-stop service action. -1 = will not perform, 0 = completed, 1 = in progress.

Type	Field	Meaning
int8_t	damage	Body-repair action state
int8_t	fuel	Refuelling action state
int8_t	tyres_lf	Front-left tyre change state
int8_t	tyres_rf	Front-right tyre change state
int8_t	tyres_lr	Rear-left tyre change state
int8_t	tyres_rr	Rear-right tyre change state

SMEvoElectronics [128 bytes]

All driver-adjustable electronic aid and setup settings.

Type	Field	Meaning
int8_t	tc_level	Traction-control level (0 = off, higher = more aggressive)
int8_t	tc_cut_level	TC throttle-cut aggressiveness level
int8_t	abs_level	ABS intervention level (0 = off)
int8_t	esc_level	Electronic stability-control level (0 = off)
int8_t	ebb_level	Electronic brake-balance adjustment level
float	brake_bias	Front brake-bias ratio (e.g. 0.56 = 56 % front)
int8_t	engine_map_level	Engine map / power mode selection
float	turbo_level	Turbo wastegate or boost target setting
int8_t	ers_deployment_map	ERS power-deployment strategy map
float	ers_recharge_map	ERS recharge aggressiveness setting
bool	is_ers_heat_charging_on	ERS heat-based charging is enabled
bool	is_ers_overtake_mode_on	ERS overtake (maximum-deploy) mode is active
bool	is_drs_open	DRS flap is currently open
int8_t	diff_power_level	Differential lock level under power
int8_t	diff_coast_level	Differential lock level on lift / coast
int8_t	front_bump_damper_level	Front bump (compression) damper stiffness level
int8_t	front_rebound_damper_level	Front rebound damper stiffness level
int8_t	rear_bump_damper_level	Rear bump (compression) damper stiffness level
int8_t	rear_rebound_damper_level	Rear rebound damper stiffness level
bool	is_ignition_on	Ignition switch is on
bool	is_pitlimiter_on	Pit-speed limiter is active
int8_t	active_performance_mode	Selected vehicle performance / power mode index

SMEvoInstrumentation [128 bytes]

Cockpit light, display, and instrumentation panel states.

Type	Field	Meaning
int8_t	main_light_stage	Main exterior light stage (0 = off)
int8_t	special_light_stage	Auxiliary / special lights level
int8_t	cockpit_light_stage	Interior cockpit illumination level
int8_t	wiper_level	Windscreen wiper speed (0 = off)
bool	rain_lights	Rear rain light is on

Type	Field	Meaning
bool	direction_light_left	Left turn indicator is active
bool	direction_light_right	Right turn indicator is active
bool	flashing_lights	Flashing lights are active
bool	warning_lights	Hazard lights are illuminated
int8_t	selected_display_index	Index of the currently focused display device
int8_t	display_current_page_index[16]	Active page index on displays
bool	are_headlights_visible	Headlights are on and visible to other drivers

SMEvoSessionState [256 bytes]

Server-side session lifecycle information.

Type	Field	Meaning
char[33]	phase_name	Name of the current session phase (e.g. 'Race', 'Qualify')
char[15]	time_left	Formatted remaining session time (HH:MM:SS)
int32_t	time_left_ms	Remaining session time in milliseconds
char[15]	wait_time	Formatted wait time before session start
int32_t	total_lap	Total laps scheduled for this session
int32_t	current_lap	Current lap number being driven
int32_t	lights_on	Number of starting lights currently illuminated
int32_t	lights_mode	Starting-light sequence mode identifier
float	lap_length_km	Track lap length in kilometres
int32_t	end_session_flag	Non-zero when the session is ending
char[15]	time_to_next_session	Formatted countdown to the next session
bool	disconnected_from_server	Player has lost connection to the game server
bool	restart_season_enabled	Season restart option is available to the player
bool	ui_enable_drive	Drive button is enabled in the UI
bool	ui_enable_setup	Setup screen is accessible from the UI
bool	is_ready_to_next_blinking	Ready-to-proceed indicator is blinking
bool	show_waiting_for_players	Waiting-for-players lobby screen is shown

SMEvoTimingState [256 bytes]

Lap timing and delta values displayed on the HUD.

Type	Field	Meaning
char[15]	current_laptime	Current lap time as a formatted string
char[15]	delta_current	Delta vs. current reference lap (formatted)
int32_t	delta_current_p	Sign of delta_current: +1 slower, -1 faster, 0 = hidden
char[15]	last_laptime	Last completed lap time as a formatted string
char[15]	delta_last	Delta vs. last lap (formatted)
int32_t	delta_last_p	Sign of delta_last: +1 slower, -1 faster, 0 = hidden
char[15]	best_laptime	Personal best lap time as a formatted string
char[15]	ideal_laptime	Theoretical best lap (sum of best sectors) as a formatted string
char[15]	total_time	Total elapsed session time as a formatted string
bool	is_invalid	Current lap has been invalidated (track-limits violation, etc.)

SMEvoAssistsState [64 bytes]

Driver-assist settings currently active for the player car.

Type	Field	Meaning
uint8_t	auto_gear	Automatic gearshift aid level (0 = off)
uint8_t	auto_blip	Automatic throttle blip on downshift (0 = off)
uint8_t	auto_clutch	Automatic clutch management (0 = off)
uint8_t	auto_clutch_on_start	Automatic clutch during the rolling start (0 = off)
uint8_t	manual_ignition_e_start	Manual ignition and electric start required (0 = automatic)
uint8_t	auto_pit_limiter	Pit-speed limiter activates automatically (0 = manual)
uint8_t	standing_start_assist	Standing-start launch assistance active (0 = off)
float	auto_steer	Auto-steer correction strength (0.0 = off, 1.0 = maximum)
float	arcade_stability_control	Arcade-style stability aid level (0.0 = off, 1.0 = maximum)

SPageFileGraphicEvo

Main HUD and graphics telemetry page. Updated each rendered frame. Contains embedded sub-structs for tyres, damage, electronics, timing, and session state.

Type	Field	Meaning
int	packetId	Incrementing counter — detect new frames by comparing to previous value
ACEVO_STATUS	status	Current simulator operational state (see ACEVO_STATUS)
uint64_t	focused_car_id_a	Unique ID of the car currently shown by the camera
uint64_t	focused_car_id_b	
uint64_t	player_car_id_a	Unique ID of the player's own car
uint64_t	player_car_id_b	
unsigned short	rpm	Engine speed in RPM for HUD display
bool	is_rpm_limiter_on	Rev limiter is cutting fuel / ignition (bouncing off limiter)
bool	is_change_up_rpm	Engine RPM is in the upshift window
bool	is_change_down_rpm	Engine RPM is in the downshift window
bool	tc_active	Traction control is actively intervening this frame
bool	abs_active	ABS is actively modulating brake pressure this frame
bool	esc_active	Electronic stability control is intervening this frame
bool	launch_active	Launch control system is engaged
bool	is_ignition_on	Ignition switch is on
bool	is_engine_running	Engine is running
bool	kers_is_charging	KERS/ERS battery is currently being charged
bool	is_wrong_way	Car is travelling in the wrong direction on track
bool	is_drs_available	DRS activation is permitted in this section
bool	battery_is_charging	High-voltage battery pack is in charging state
bool	is_max_kj_per_lap_reached	Maximum ERS deployment energy for this lap has been consumed
bool	is_max_charge_kj_per_lap_reached	Maximum ERS charge energy for this lap has been stored
short	display_speed_kmh	Displayed speed in km/h
short	display_speed_mph	Displayed speed in mph
short	display_speed_ms	Displayed speed in m/s
float	pitspeeding_delta	Speed delta vs. pit-lane limit (negative = under limit)
short	gear_int	Current gear as an integer (same encoding as physics gear)
float	rpm_percent	Engine RPM as a fraction of redline (0.0–1.0)
float	gas_percent	Throttle pedal position as a fraction (0.0–1.0)
float	brake_percent	Brake pressure as a fraction (0.0–1.0)
float	handbrake_percent	Handbrake engagement as a fraction (0.0–1.0)
float	clutch_percent	Clutch disengagement as a fraction (1.0–0.0)
float	steering_percent	Steering wheel position (–1.0 = full left, +1.0 = full right)
float	ffb_strength	Global force-feedback output strength
float	car_ffb_muplier	Per-car force-feedback gain multiplier
float	water_temperature_percent	Coolant temperature as a fraction of optimal operating range
float	water_pressure_bar	Coolant system pressure in bar

Type	Field	Meaning
float	<code>fuel_pressure_bar</code>	Fuel system pressure in bar
int8_t	<code>water_temperature_c</code>	Coolant temperature in °C
int8_t	<code>air_temperature_c</code>	Ambient air temperature in °C
float	<code>oil_temperature_c</code>	Engine oil temperature in °C
float	<code>oil_pressure_bar</code>	Engine oil pressure in bar
float	<code>exhaust_temperature_c</code>	Exhaust gas temperature in °C
float	<code>g_forces_x</code>	Lateral G-force (positive = rightward)
float	<code>g_forces_y</code>	Longitudinal G-force (positive = under acceleration)
float	<code>g_forces_z</code>	Vertical G-force (positive = upward)
float	<code>turbo_boost</code>	Absolute turbo boost pressure in bar
float	<code>turbo_boost_level</code>	Current boost stage or map level
float	<code>turbo_boost_perc</code>	Turbo boost as a fraction of maximum (0.0–1.0)
int32_t	<code>steer_degrees</code>	Steering wheel rotation in degrees from centre
float	<code>current_km</code>	Distance driven in the current session in km
uint32_t	<code>total_km</code>	Total odometer / career distance in km
uint32_t	<code>total_driving_time_s</code>	Total driving time accumulated in seconds
int32_t	<code>time_of_day_hours</code>	In-game time of day — hours (0–23)
int32_t	<code>time_of_day_minutes</code>	In-game time of day — minutes (0–59)
int32_t	<code>time_of_day_seconds</code>	In-game time of day — seconds (0–59)
int32_t	<code>delta_time_ms</code>	Delta vs. reference lap in milliseconds (signed)
int32_t	<code>current_lap_time_ms</code>	Current lap time in milliseconds
int32_t	<code>predicted_lap_time_ms</code>	Predicted final lap time in milliseconds
float	<code>fuel_liter_current_quantity</code>	Fuel remaining in the tank in litres
float	<code>fuel_liter_current_quantity_percent</code>	Fuel remaining as a fraction of tank capacity
float	<code>fuel_liter_per_km</code>	Average fuel consumption rate in litres per km
float	<code>km_per_fuel_liter</code>	Average fuel economy in km per litre
float	<code>current_torque</code>	Engine output torque in Nm
int32_t	<code>current_bhp</code>	Engine output power in brake horsepower
SMEvoTyreState	<code>tyre_lf</code>	Full tyre state for the front-left corner
SMEvoTyreState	<code>tyre_rf</code>	Full tyre state for the front-right corner
SMEvoTyreState	<code>tyre_lr</code>	Full tyre state for the rear-left corner
SMEvoTyreState	<code>tyre_rr</code>	Full tyre state for the rear-right corner
float	<code>npos</code>	Normalised track position (0.0 = start/finish line, 1.0 = one full lap)
float	<code>kers_charge_perc</code>	KERS/ERS charge level as a fraction (0.0–1.0)
float	<code>kers_current_perc</code>	KERS/ERS power currently being deployed as a fraction
float	<code>control_lock_time</code>	Seconds driver input remains locked (e.g. after collision penalty)
SMEvoDamageState	<code>car_damage</code>	Damage levels for each body zone of the car
ACEVO_CAR_LOCATION	<code>car_location</code>	Current track zone the car occupies (see ACEVO_CAR_LOCATION)
SMEvoPitInfo	<code>pit_info</code>	Status of each pit-stop service item
float	<code>fuel_liter_used</code>	Fuel consumed since session start in litres

Type	Field	Meaning
float	fuel_liter_per_lap	Average fuel consumed per lap in litres
float	laps_possible_with_fuel	Estimated number of laps achievable with remaining fuel
float	battery_temperature	High-voltage battery temperature in °C
float	battery_voltage	High-voltage battery pack voltage in V
float	instantaneous_fuel_liter_per_km	Instantaneous fuel consumption in litres per km
float	instantaneous_km_per_fuel_liter	Instantaneous fuel economy in km per litre
float	gear_rpm_window	How well current RPM suits the engaged gear (1.0 = ideal window)
SMEvoInstrumentation	instrumentation	Current state of all cockpit lights and displays
SMEvoInstrumentation	instrumentation_min_limit	Minimum allowed setting for each instrumentation item
SMEvoInstrumentation	instrumentation_max_limit	Maximum allowed setting for each instrumentation item
SMEvoElectronics	electronics	Current electronic aid and setup values
SMEvoElectronics	electronics_min_limit	Minimum allowed value for each electronics setting
SMEvoElectronics	electronics_max_limit	Maximum allowed value for each electronics setting
SMEvoElectronics	electronics_is_modifiable	Flags which electronics fields the driver can adjust in-session
int32_t	total_lap_count	Total laps completed in the session
uint32_t	current_pos	Current race position (1 = leader)
uint32_t	total_drivers	Total number of cars in the session
int32_t	last_laptime_ms	Last completed lap time in milliseconds
int32_t	best_laptime_ms	Personal best lap time in milliseconds
ACEVO_FLAG_TYPE	flag	Flag shown specifically to this driver
ACEVO_FLAG_TYPE	global_flag	Flag shown to all drivers on track
uint32_t	max_gears	Number of forward gears the car has
ACEVO_ENGINE_TYPE	engine_type	Powertrain type of the car (see ACEVO_ENGINE_TYPE)
bool	has_kers	Car is equipped with a KERS/ERS system
bool	is_last_lap	This is the final scheduled lap of the race
char[33]	performance_mode_name	Display name of the active vehicle performance / power mode
float	diff_coast_raw_value	Raw differential coast-lock value from setup
float	diff_power_raw_value	Raw differential power-lock value from setup
int32_t	race_cut_gained_time_ms	Cumulative time penalty from track-limit cuts in ms
int32_t	distance_to_deadline	Distance to the penalty trigger in metres
float	race_cut_current_delta	Running delta time accrued from track-limit violations
SMEvoSessionState	session_state	Session lifecycle and countdown information
SMEvoTimingState	timing_state	HUD lap times and delta display values
int32_t	player_ping	Network round-trip ping to the server in ms
int32_t	player_latency	Measured network latency in ms
int32_t	player_cpu_usage	Client CPU usage in percent
int32_t	player_cpu_usage_avg	Average client CPU usage in percent
int32_t	player_qos	Network Quality-of-Service score
int32_t	player_qos_avg	Average QoS score over the session

Type	Field	Meaning
int32_t	player_fps	Current rendered frames per second
int32_t	player_fps_avg	Average FPS over the session
char[33]	driver_name	Driver's first name
char[33]	driver_surname	Driver's surname
char[33]	car_model	Identifier or display name of the car model
bool	is_in_pit_box	Car is stationary inside its assigned pit box
bool	is_in_pit_lane	Car is anywhere within the pit lane
bool	is_valid_lap	Current lap is valid and counts for timing
float[60][3]	car_coordinates	World-space position of up to 60 cars [car_index][X, Y, Z]
float	gap_ahead	Time gap to the car immediately ahead in seconds
float	gap_behind	Time gap to the car immediately behind in seconds
uint8_t	active_cars	Number of cars actively participating in the session
float	fuel_per_lap	Target fuel consumption per lap in litres
float	fuel_estimated_laps	Estimated laps remaining with current fuel
SMEvoAssistsState	assists_state	All driver-assist levels currently active
float	max_fuel	Maximum fuel tank capacity of the car in litres
float	max_turbo_boost	Maximum turbo boost pressure in bar
bool	use_single_compound	Car is restricted to a single tyre compound for both axles

SPageFileStaticEvo

Static session metadata. Written once when a session loads and does not change while driving.

Type	Field	Meaning
char[15]	sm_version	Shared-memory interface version string
char[15]	ac_evo_version	AC Evo game build version string
ACEVO_SESSION_TYPE	session	Type of the current session (see ACEVO_SESSION_TYPE)
char[33]	session_name	Human-readable session name (e.g. 'Race 1')
uint8_t	event_id	Unique identifier of the event within the championship
uint8_t	session_id	Unique identifier of this session within the event
ACEVO_STARTING_GRIP	starting_grip	Tyre grip condition at session start (see ACEVO_STARTING_GRIP)
float	starting_ambient_temperature_c	Ambient air temperature at session start in °C
float	starting_ground_temperature_c	Road surface temperature at session start in °C
bool	is_static_weather	Weather is fixed and will not change during the session
bool	is_timed_race	Session ends by elapsed time rather than lap count
bool	is_online	Session is an online multiplayer event
int	number_of_sessions	Total sessions in this event (e.g. 3 = practice + qualify + race)
char[33]	nation	Country / nation name associated with the event or track
float	longitude	Geographic longitude of the track location in decimal degrees
float	latitude	Geographic latitude of the track location in decimal degrees
char[33]	track	Track identifier or name
char[33]	track_configuration	Track layout variant or configuration name
float	track_length_m	Total lap length of the track in metres